

Tutorial

Modern Physics

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For exercises from the textbook, the corresponding number is reported. Text may have been edited, for clarity.
Unless otherwise specified, all results are given with three significant digits, avoiding rounding errors.

The solutions are provided on an "as is" basis, without warranty. Any error report is appreciated.

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WEEK 10

SPECIAL RELATIVITY

• Simple calculations / Ex. 2.12

Determine the ratio $\beta = v/c$ for

- (a) a car traveling 95 km/h
- (b) a commercial jet airliner traveling 240 m/s
- (c) a supersonic airplane traveling at Mach 2.3; Mach number = v/v_{sound} , $v_{\text{sound}} = 330$ m/s
- (d) a space station, traveling 27 000 km/h
- (e) an electron traveling 25 cm in 2 ns
- (f) a proton traveling across a nucleus (10^{-14} m) in 0.35×10^{-22} s

Solution

- (a) 8.80×10^{-8}
- (b) 8.00×10^{-7}
- (c) 2.53×10^{-6}
- (d) 2.50×10^{-5}
- (e) 0.417
- (f) 0.952

• Tricky calculations / Ex. 2.15

Find the relativistic factor $\gamma = 1/\sqrt{1 - \beta^2}$ for each of the previous entries.

Comment

It is worth noticing that for $v \ll c$ even a scientific calculator will give an (approximate) answer $\gamma = 1$ because the deviation from unity is below the precision of floating point numbers. In these cases, the suggested strategy is to use the binomial expansion

$$\gamma = (1 - \beta^2)^{-1/2} \approx 1 + \frac{\beta^2}{2}$$

and give the result in the form $\gamma \approx 1 + \epsilon$.

Solution

- (a) $\gamma \approx 1 + \beta^2/2 = 1 + 3.87 \times 10^{-15}$
- (b) $\gamma \approx 1 + \beta^2/2 = 1 + 3.20 \times 10^{-13}$
- (c) $\gamma \approx 1 + \beta^2/2 = 1 + 3.10 \times 10^{-12}$
- (d) $\gamma \approx 1 + \beta^2/2 = 1 + 3.10 \times 10^{-10}$
- (e) $\gamma = 1.10$
- (f) $\gamma = 3.20$

• Lorentz transformation / Ex. 2.16

An event occurs in system K' at $x' = 2$ m, $y' = 3.5$ m, $z' = 3.5$ m and $t' = 0$. System K' and K have their axes coincident at $t = t' = 0$, and system K' travels along the x axis of system K with a speed $v = 0.8c$. What are the coordinates of the event in system K ?

Solution

To find the coordinates we use the inverse Lorentz's transformation (with $\beta = v/c = 0.8$)

$$\begin{cases} x = \gamma(x' + vt') \\ t = \gamma(t' + \beta x'/c), \end{cases} \quad (1)$$

yielding $x = 3.33$ m and $t = 8.89 \times 10^{-9}$ s.

Note that $y = y'$ and $z = z'$ because system K' is not in motion with respect to K along those directions.

• **Sending a light signal / Ex 2.17**

A light signal is sent from the origin of a system K $(x_1, y_1, z_1) = (0, 0, 0)$ at $t_1 = 0$ to the point $(x_2, y_2, z_2) = (3, 5, 10)$ m. (a) At what time t_2 is the signal received? (b) Find (x'_2, y'_2, z'_2) and t'_2 for the receipt of the signal in a frame K' that is moving along the x axis of K at a speed of $v = 0.8c$. (c) From your results in (b), verify that the light traveled with a speed c as measured in the K' frame.

Solution

For a *lightlike* event the invariant spacetime interval vanishes:

$$\Delta s^2 = \Delta x^2 + \Delta y^2 + \Delta z^2 - c^2 \Delta t^2 = 0,$$

from which one can compute $\Delta t^2 = t_2 - t_1 = t_2 = 3.86 \times 10^{-8}$ second.

Since K' moves only along the x direction, the Lorentz's transformation yield $y'_2 = y_2$ and $z'_2 = z_2$, whereas

$$\begin{cases} x'_2 = \gamma(x_2 - vt_2) \\ t'_2 = \gamma(t_2 - \beta x_2/c), \end{cases} \quad (2)$$

which, after substituting the values for x_2 and t_2 , yields $x' = -10.4$ m and $t' = 51.0$ ns.

Since the origin of K and K' frame coincide at $t = t' = 0$, the velocity of the signal is given by

$$v = (x'_2 + y'_2 + z'_2)^{1/2} / t'_2 = 3.00 \times 10^8 \text{ m/s}$$

which corresponds to the speed of light in the vacuum.

Note that if we substitute the values above instead of carrying all digits, one finds an approximate (and suboptimal) solution $v = 2.994$ m/s instead.

• **Experimental verification of special relativity / Ex 2.25**

The Concorde traveled a distance $\ell = 8000$ km between two places in North America and Europe (as naturally measured in the reference frame at rest on Earth) at an average speed of $v = 375$ m/s. What is the total difference in time Δt between two similar atomic clocks, one on the airplane (t_0) and one at rest on Earth (t) during a one-way trip? Consider only time dilation and ignore other effects such as Earth's rotation.

Solution

Accounting for time dilation, the proper time t_0 measured in the co-moving frame of the clock on the airplane is given by $t_0 = t/\gamma$, and the difference in time is given by

$$\Delta t = t - t_0 = t(1 - \gamma^{-1}) > 0$$

with

$$t = \frac{\ell}{v}.$$

Considering that $v \ll c$, it is convenient to approximate $\gamma^{-1} = \sqrt{1 - \beta^2} \approx 1 - \beta^2/2$. Hence, the extra time measured by the atomic clock on Earth is given by

$$\Delta t \approx \frac{\ell}{v} \frac{\beta^2}{2} = 16.7 \text{ ns.}$$

Note that when considering time dilation, we always have a clock that is at rest in a moving frame (the clock on the plane) and several synchronized clocks in another ("rest") frame. In the formula for proper time it is always the moving clock that measures shorter time compared to the stationary clocks. This is not in contradiction with relativity, because in this experiment we can tell apart the "rest" frame by the need of using more than one clock, one in Europe and one in America.

• **Shooting at satellites / Ex. 2.26**

A mechanism on Earth used to shoot down geosynchronous satellites that house laser-based weapons is finally perfected and propels golf balls at $v = 0.94c$. Geosynchronous satellites are placed at $\ell = 3.58 \times 10^4$ km from the surface of the Earth (i.e., measured in a reference frame at rest on Earth). (a) What is the distance ℓ' from the Earth to the satellite, as measured by a detector placed inside the golf ball? (b) How much time t will it take the golf ball to make the journey to the satellite in the Earth's frame? How much time t' will it take in the golf ball's frame?

Solution

The distance measured in the frame co-moving with the golf ball is given by

$$\ell' = \frac{\ell}{\gamma} = 1.22 \times 10^7 \text{ m.}$$

The time measured in the Earth's frame is then

$$t = \frac{\ell}{v} = 0.127 \text{ s,}$$

and the time in the ball's frame is

$$t' = \frac{t}{\gamma} = 0.0433 \text{ s.}$$

• **Muon decay / Ex. 2.21**

Particle physicists use particle track detectors to determine the lifetime of short-lived particles. A muon has a mean lifetime of $t' = 2.2 \mu\text{s}$ (as measured in the reference frame co-moving with the muon) and makes a track $\ell = 9.5 \text{ cm}$ long (as measured in the lab's frame) before decaying into an electron and two neutrinos. What was the speed v of the muon?

Solution

In the lab frame, the lifetime of the muon is longer, due to time dilation

$$t = \frac{\ell}{v} = \gamma t' > t'.$$

Rearranging the terms and using $v = \beta c$ one obtains

$$\frac{t'}{\sqrt{1 - \beta^2}} = \frac{\ell}{\beta c}$$

and solving for β and then obtaining v yields

$$v = c \left[1 + \left(\frac{ct'}{\ell} \right)^2 \right]^{-1/2} = 1.44 \times 10^{-4} c.$$

• **Velocity addition / Ex 2.93**

A spaceship moving with $v = 0.80c$ in a direction away from Earth (e.g., x) fires a missile that the spaceship measures to be moving at $u' = 0.80c$ perpendicular to the ship's direction of travel (e.g., y). Find the velocity components $u_{x,y,z}$ and speed u of the missile as measured by Earth.

Solution

The two relevant coordinates are x and y . Making the components explicit, one has $v_x = 0.8c$ and $v_y = 0$, while $u'_x = 0$ and $u'_y = u'$. Using the velocity addition formula, one finds

$$u_x = \frac{u'_x + v_x}{1 + \left(\frac{v_x}{c^2} \right) u'_x} = 0.8 c$$

and

$$u_y = \frac{u'_y + v_y}{\gamma \left[1 + \left(\frac{v_y}{c^2} \right) u'_x \right]} = 0.48 c.$$

Note that the denominator has u'_x , in the tutorial I made the mistake of writing u'_y instead.